

INTRODUCTION TO GEOPHYSICAL PROSPECTING

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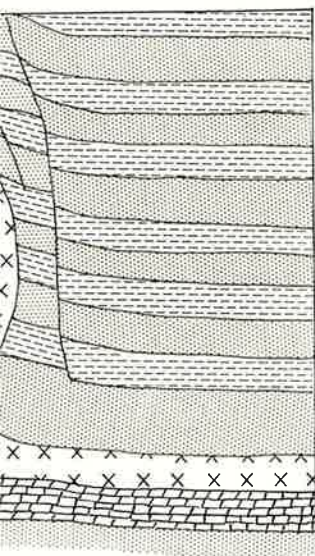
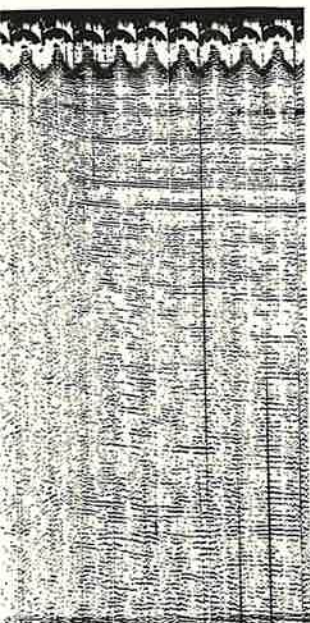
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salt dome in Louisiana. Geologi-
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12-2. Choice of Density

In the interpretation of gravity anomalies, it is necessary to estimate the densities of the subsurface rocks before one can postulate their structure. For this reason it is desirable to give some data on the densities of representative rocks in regions where gravity surveys are ordinarily made. As pointed out in Chap. 8, it is not the absolute densities but the density contrasts that are significant. Among sedimentary rocks these contrasts are almost always small, seldom exceeding 0.25 g/cm³. Moreover, the variation in the possible densities of almost any type of rock results in

TABLE 12-1. Densities of Typical Rock Materials*

Sedimentary Rocks		Density, g/cm ³	
Character and age of formation	Location	Dry	Water saturated
Arenaceous rocks:			
Recent alluvium.....	St. Charles Co., Mo.	1.53-1.55	1.96-1.97
Loess.....	Fremont Co., Idaho	1.01	1.62
Recent silt.....	Rosebud Co., Mont.	1.36	1.86
Kirkwood sand, Miocene.....	Margate City, N.J.	1.56	1.97
Fort Union sand, Tertiary.....	Rosebud Co., Mont.	1.79-1.81	2.14
Berea sandstone, Mississippian.....	Ohio	2.23-2.54	2.40-2.59
Domengine sandstone, Eocene.....	North Belridge, Calif.	2.56-2.63	2.67-2.73
Dakota sandstone, Cretaceous.....	Lewis, Iowa	1.65-1.76	2.03-2.10
St. Peter sandstone, Ordovician.....	Ozark Plateau, Okla.	2.28-2.55	2.42-2.59
Argillites:			
White Flint clay, Pennsylvanian.....	Fulton, Mo.	2.37	2.47
Middendorf clay, Cretaceous.....	Richland Co., S.C.	1.51	1.93
Hamilton shale, Devonian.....	Hannibal, Mo.	2.32	2.43-2.44
Nonesuch shale, pre-Cambrian.....	Michigan	2.80	
Carlyle shale, Cretaceous.....	Weston Co., Wyo.	2.00	2.24
Bearpaw shale, Cretaceous.....	Rosebud Co., Mont.	1.57	1.98
Carbonates:			
Greenhorn limestone, Cretaceous.....	Crook Co., Wyo.	1.74	2.12
Chalk.....	England	1.53-2.22	1.96-2.40
Oolitic limestone.....	England	2.16	2.36
Dolomite.....	England	2.54	2.63
Marble, 34 samples.....	Twelve states	2.66-2.86	2.68-2.86
Evaporites, etc.:			
Anhydrite.....		2.9	
Gypsum.....		2.2	
Chert.....	England	2.29	2.40
Diatomite.....	Ontario, Oreg.	0.45	1.18
Rock salt.....		2.2	

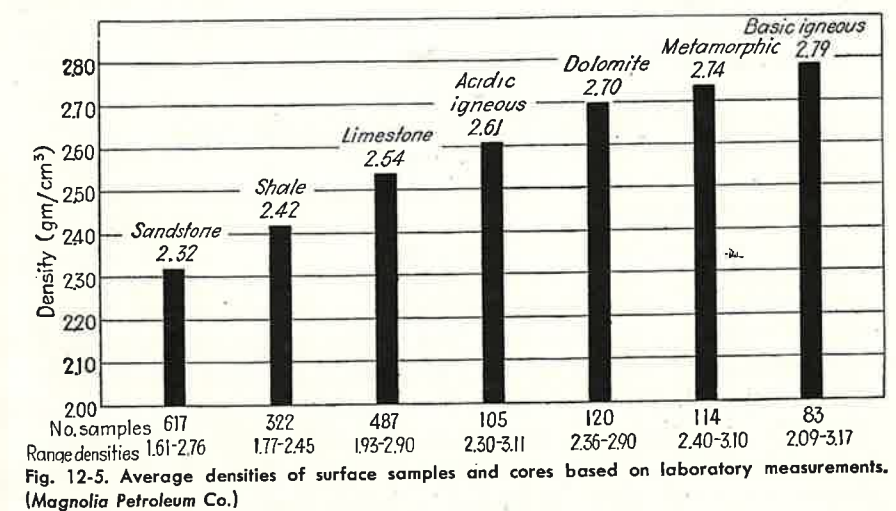
TABLE 12-1. Densities of Typical Rock Materials* (Continued)

Holocrystalline Igneous Rocks		
Rock	Number of samples	Range of densities, g/cm ³
Granite.....	155	2.516-2.809
Granodiorite.....	11	2.668-2.785
Syenite.....	24	2.630-2.899
Diorite.....	13	2.721-2.960
Norite.....	11	2.720-3.020
Gabbro.....	27	2.850-3.120
Diabase.....	40	2.804-3.110
Peridotite.....	3	3.152-3.276
Dunite.....	1	3.289
Pyroxenite.....	8	3.10-3.318
Anorthosite.....	12	2.640-2.920

* Values taken from "Handbook of Physical Constants," edited by Francis Birch, Geological Society of America Special Paper 36, 1942.

some uncertainty as to the proper value to use in actual computations. Table 12-1 illustrates the range of density values encountered in a number of typical rock materials as well as the differences that might be expected between the different types.

It is evident from this table that each type of sedimentary rock covers a wide range of densities and that the ranges are quite similar for sandstones, shales, and limestones. Igneous rocks tend to have higher density than sedimentary formations. Even here, however, there is substantial



10⁻⁵ oersted, is more convenient and has become the most commonly used unit of field intensity for geophysical work.

Electromagnetism. Every electric current generates a magnetic field which is in a plane perpendicular to the current, as shown by the orientation of the compass needles around the wire in Fig. 13-4. The strength of the field is proportional to the current and in the case of a long, straight

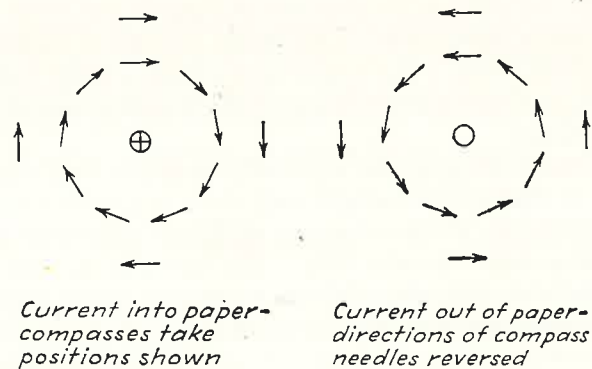


Fig. 13-4. Orientation of compass needles around straight wire (perpendicular to paper) carrying current. This experiment shows that current creates magnetic field with circular lines of force having their center at wire.

wire is inversely proportional to the distance from the wire. This principle is important in magnetic prospecting only in so far as it forms the basis for certain types of geomagnetic instruments.

13-2. Magnetic Susceptibility of Rocks

The most significant magnetic property of rocks is their susceptibility. This can be measured with the rocks in place by an induction balance developed by Mooney.³ Pulverized samples of rocks can be placed near a field magnetometer and the deflection of the magnetometer needle caused by the sample can be used to calculate the susceptibility. *In situ* measurements are preferable wherever it is feasible to make them.

In the laboratory, primary and secondary coils are wound about the rock sample whose susceptibility is to be measured. The deflection of a galvanometer connected to the secondary coil is noted when a current is sent through the primary. Barret⁴ describes an apparatus based on this principle which is designed primarily for rock-susceptibility measurements.

When an external magnetic field is used for measuring susceptibility, it is customary to specify the strength of this field when the results are tabulated. The polarization should properly consist of two parts, the susceptibility polarization kH , dependent on the external field H , and

the remanent polarization I_p , which governs the residual magnetism in the absence of such a field.

Table 13-1 gives some representative values of susceptibility and polarization (after Heiland and Birch) for several different kinds of rocks and minerals.

TABLE 13-1. Measured Susceptibilities of Rock Materials

Material	$k \times 10^6$, cgs	At H , oersteds
Magnetite	300,000-800,000	0.6
Pyrrhotite	125,000	0.5
Ilmenite	135,000	1
Franklinite	36,000	
Dolomite	14	0.5
Sandstone	16.8	1
Serpentine	14,000	30.5
Granite	28-2,700	1
Diorite	46.8	1
Gabbro	68.1-2,370	1
Porphyry	47	1
Diabase	78-1,050	1
Basalt	680	1
Olivine-diabase	2,000	0.5
Peridotite	12,500	0.5-1.0

TABLE 13-2. Calculated Susceptibilities of Rock Materials

Material	Magnetite						Ilmenite	
	Minimum		Maximum		Average		Average	
	Per cent	$k \times 10^6$	Per cent	$k \times 10^6$	Per cent	$k \times 10^6$	Per cent	$k \times 10^6$
Quartz porphyries	0.0	0	1.4	4,200	0.82	2,500	0.3	410
Rhyolites	0.2	600	1.9	5,700	1.00	3,000	0.45	610
Granites	0.2	600	1.9	5,700	0.90	2,700	0.7	1,000
Trachyte-syenites	0.0	0	4.6	14,000	2.04	6,100	0.7	1,000
Eruptive nephelites	0.0	0	4.9	15,000	1.51	4,530	1.24	1,700
Abyssal nephelites	0.0	0	6.6	20,000	2.71	8,100	0.85	1,100
Pyroxenites	0.9	3,000	8.4	25,000	3.51	10,500	0.40	5,400
Gabbros	0.9	3,000	3.9	12,000	2.40	7,200	1.76	2,400
Monzonite-latites	1.4	4,200	5.6	17,000	3.58	10,700	1.60	2,200
Leucite rocks	0.0	0	7.4	22,000	3.27	9,800	1.94	2,600
Dacite-quartz-diorite	1.6	4,800	8.0	24,000	3.48	10,400	1.94	2,600
Andesites	2.6	7,800	5.8	17,000	4.50	13,500	1.16	1,600
Diorites	1.2	3,600	7.4	22,000	3.45	10,400	2.44	4,200
Peridotites	1.6	4,800	7.2	22,000	4.60	13,800	1.31	1,800
Basalts	2.3	6,900	8.6	26,000	4.76	14,300	1.91	2,600
Diabases	2.3	6,900	6.3	19,000	4.35	13,100	2.70	3,600

The majority of rock-susceptibility values to be found in tables were measured in fields greater than the earth's. It is questionable how useful such values are for geophysical prospecting in view of the fact that the susceptibility as measured in a strong field is lower than when determined in a field of $\frac{1}{2}$ oersted or thereabouts. For prospecting calculations, susceptibilities determined according to a somewhat indirect method

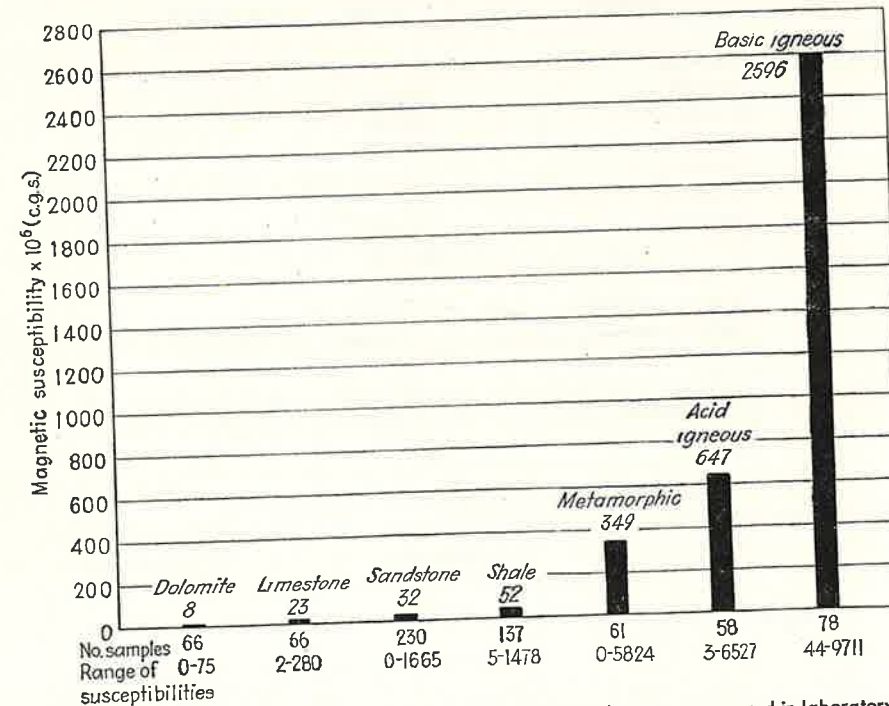


Fig. 13-5. Average magnetic susceptibilities of surface samples and cores as measured in laboratory. (Magnolia Petroleum Co.)

proposed by Slichter⁵ are probably safer. Assuming that the magnetism of most rocks is attributable to their magnetite content, he obtains the susceptibility of the rock by multiplying the volume percentage of the magnetite in the rock by the susceptibility of magnetite (considered as 0.3 cgs). He finds good agreement between values calculated in this manner and those measured directly in fields having the same strength as the earth's. Stearn⁶ has tabulated the magnetite and ilmenite content of a large number of igneous rocks. Susceptibilities calculated by Slichter's method are presented in Table 13-2 for both constituents on the basis of Stearn's data. This table shows the range of variation in susceptibility for any given type of rock. One sees that the magnetiza-

tion attributable to the ilmenite is almost always small compared to that from the magnetite.

The Magnolia Petroleum Co. has conducted laboratory measurements of susceptibility on a large number of rock samples, igneous, metamorphic, and sedimentary. J. W. Peters has prepared a bar graph (Fig. 13-5) showing susceptibilities for a number of rock types. Although igneous and metamorphic rocks generally have higher susceptibilities than sedimentary rocks, the range of variation is so great that it is not possible to identify even the type of rock from magnetic information alone. In general, however, the magnetization of sedimentary rocks is so small that structural features confined to the sedimentary section alone will seldom if ever be reflected in magnetic profiles. For this reason, the chief value of magnetic methods for petroleum exploration is in areas where the basement rocks govern sedimentary structure. In prospecting for ores by magnetic means it is not usually possible to detect minerals other than magnetite, pyrrhotite, or ilmenite, although one instance has been reported from Switzerland where jacobsonite, $(\text{Mn, Fe})_3\text{O}_4$, was found responsible for an anomaly of more than 1,000 gammas.

13-3. Magnetic Effects from Buried Magnetic Bodies

Analytical Methods of Computation. The magnetic effects on the earth's surface that should result from buried uniformly magnetized bodies of a specified shape can be calculated from potential theory by methods similar to those used for determining the gravitational effects from the same bodies. The computations in the magnetic case are considerably more difficult, however, since the dipoles distributed through the body have both attraction and repulsion. The gravitational case is simpler because all elements of mass attract. The direction of magnetization introduces another complication. Many of the formulas for generalized geometric forms such as spheres can be derived from the corresponding gravitational formulas, making use of a convenient relationship between magnetic and gravitational potentials first demonstrated by Poisson.

Vertical Intensities from Vertically Polarized Bodies. We shall first consider the magnetic profile that would be observed in a horizontal plane over an isolated negative pole at depth z , as shown in Fig. 13-6. From Eq. (13-2), remembering that $V = (z/r)H$ (where H , equal to P/r^2 , is the total field from the pole), we have

$$V = \frac{Pz}{r^3} = \frac{Pz}{(x^2 + z^2)^{3/2}} \quad (13-7)$$

when a specified potential electrochemical activity with the basis for the self-potential. This gives information on electric charge, and it frequency alternating current prospecting techniques. A material is defined as the unit cube of that material having a length l and expressed by the formula

(17-1)

the ohm-centimeter. The V and the resistance R by

(17-2)

is $1/\rho$, the reciprocal of its

rock materials is enormous.

Rocks and minerals with considered good conductors; and from 10^{10} to 10^{17} resistivities for a number of

reference between resistivities of rocks. Certain minerals, resistivities (high conductivities) it possible to locate them by

particularly sands, gravels, concerned more by the electrical interstices within the formation rock material itself.

Electrical activity depends on also on the composition and in the ground water with property that governs the magnetism the rock material is in

TABLE 17-1. Resistivities of Typical Rock Materials

Material	Frequency at measurement, cps (if not d-c)	Resistivity, ohm-cm
Minerals:		
Galena.....	...	0.5-5.0
Pyrite.....	...	0.1
Magnetite.....	...	0.6-1.0
Graphite.....	...	0.03
Rock salt (impure).....	...	$3 \times 10^3 - 5 \times 10^5$
Serpentine.....	...	2×10^4
Siderite.....	...	7×10^3
Igneous rocks		
Granite.....	...	10^8
Granite.....	16	5×10^5
Diorite.....	...	10^6
Gabbro.....	...	$10^7 - 1.4 \times 10^9$
Diabase.....	...	3.1×10^5
Metamorphic rocks		
Garnet gneiss.....	...	2×10^7
Mica schist.....	16	1.3×10^6
Biotite gneiss.....	...	$10^8 - 6 \times 10^8$
Slate.....	...	$6.4 \times 10^4 - 6.5 \times 10^6$
Sedimentary rocks		
Chattanooga shale.....	50	$2 \times 10^3 - 1.4 \times 10^5$
Michigan shale.....	60	2×10^5
Calumet and Hecla conglomerates.....	60	$2 \times 10^6 - 1.3 \times 10^6$
Muschelkalk sandstone.....	16	7×10^3
Ferruginous sandstone.....	...	7×10^5
Muschelkalk limestone.....	16	1.8×10^4
Marl.....	...	7×10^3
Glacial till.....	...	5×10^4
Oil sand.....	...	$4 \times 10^2 - 2.2 \times 10^4$

Electrochemical activity has its most important practical application in potential logging. When a salt-water-saturated sand and an underlying impervious clay are both penetrated by a borehole filled with a high-resistance mud, an electromotive force E is generated which has a magnitude $K \log \rho_m / \rho_w$, where ρ_m is the resistivity of the mud, ρ_w is the resistivity of the connate water in the mud, and K is a constant depending on the constitution of the solids and fluids in contact. If the mud and water have the same resistivity, no potential is generated.

Dielectric Constant. The dielectric constant, which corresponds to permeability in magnetic materials, is a measure of the polarizability of a material in an electric field. The polarization or electrical moment per unit volume, P , is proportional to the impressed electrical field E and the